BearPi-HM Micro

Exclusive Edition

Revision B04

Table of contents

Sheet 1: BearPi-HM Micro overview (this page)

Sheet 2: 01 MP157A

Sheet 3: 02 Power

Sheet 4: 03 DDR

Sheet 5: 04 Storage

Sheet 6: 05 Type-C DRP

Sheet 7: 06 E53 Interface

Sheet 8: 07 Display

Sheet 9: 08 Wifi

Sheet 10: 09 User

Sheet 11: 10 Audio

Sheet 12: 11 S24_Interface

Sheet 13: 12 USB HOST

Sheet 14: 99 4.3' LCD_Board

框架说明

BearPi-IoT Std开发板总体上由以下功能模块构成

01 主控部分，采用STM32MP157高性能异构双核处理器

02 电源部分，采用PMIC芯片，为整个开发板供电

03 存储部分，板载512MB DDR3L

04 存储部分，板载通用TF卡接口，以及4GB eMMC

05 USB TYPE ―C电源输入及DEBUG串口

06 E53接口，遵循E53接口标准而设计，为开发者提供快捷的案例扩展方式

07 液晶显示屏接口，连接屏幕板

08 SDIO WIFI，搭载3881 WIFI模块，可实现数据的无线传输

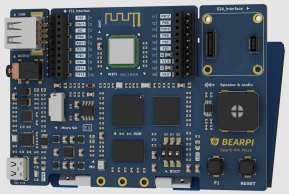
09 用户输入部分，包含两个用户按键、用户灯，以及BOOT启动模式选择

10 音频部分，搭载耳机接口，MEMS硅麦，以及高音质扬声器

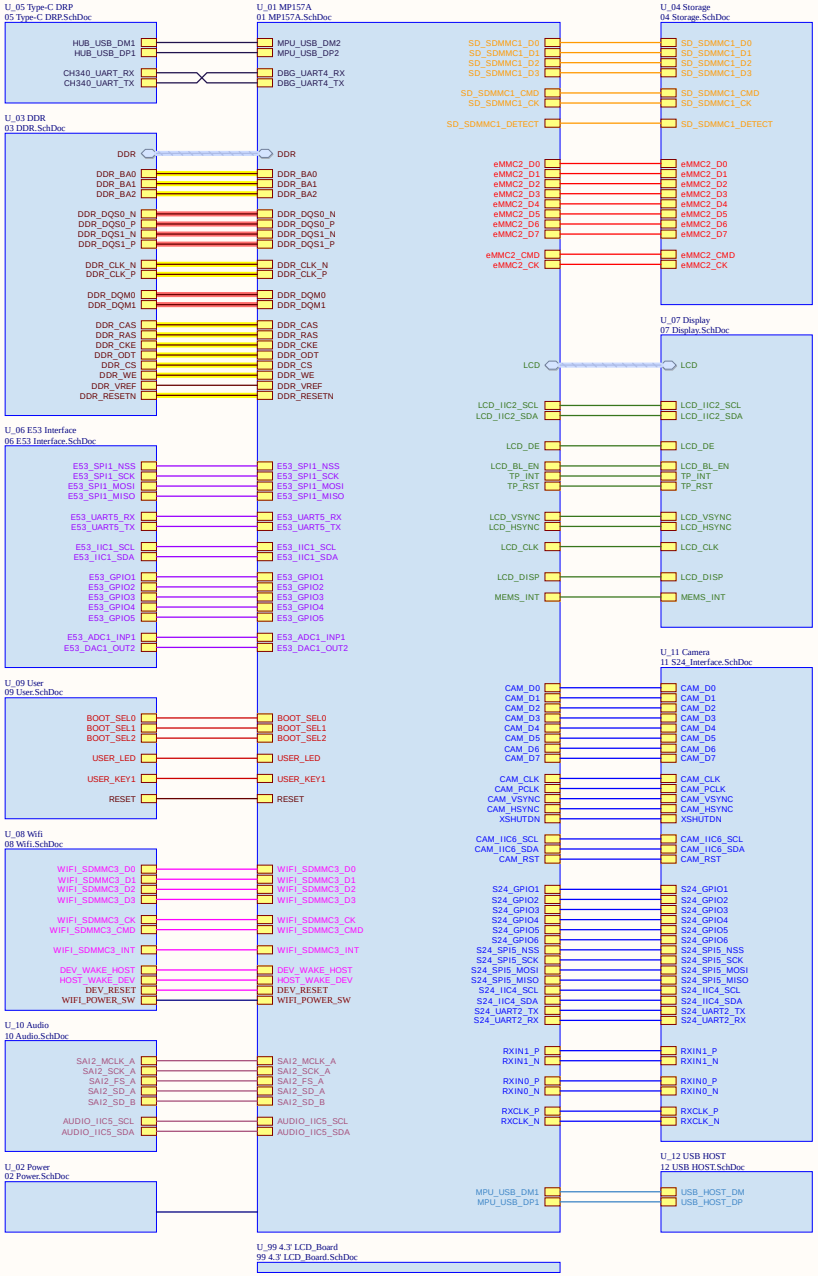
11 S24接口，遵循S24接口标准设计，可外接如高清摄像头等富设备

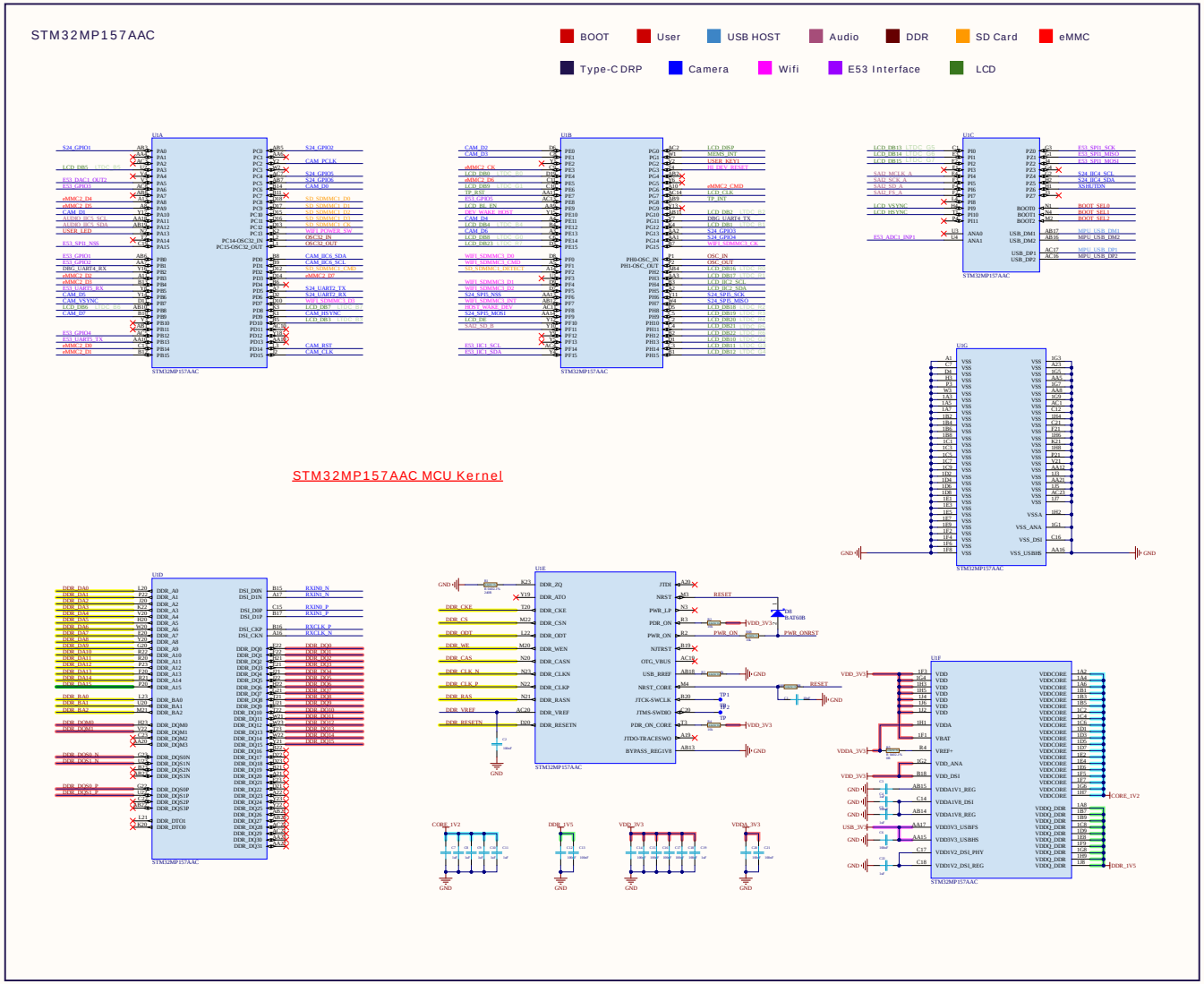
12 USB HOST部分，可外挂U盘、USB网口等USB设备

99 4.3' 液晶显示屏，搭载4.3寸TFT 液晶显示屏，为开发者提供一个直观的交互界面



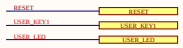
BearPi-HM Micro Overview



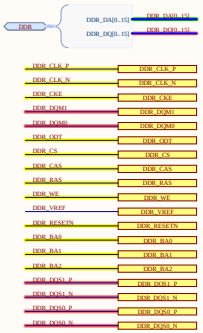


BOOT

User

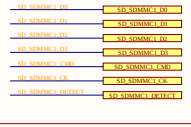


DDR

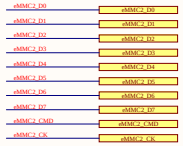


USB HOST

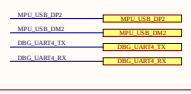
SD Card



eMMC

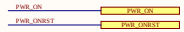


Type-C DRP

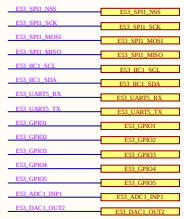


LCD

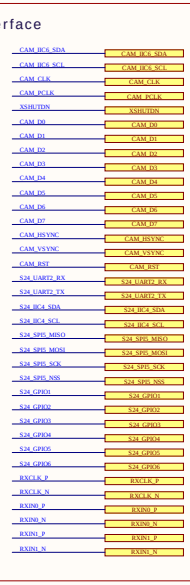
POWER



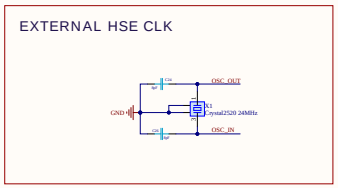
E53 Interface



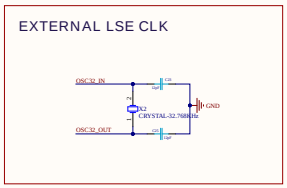
S24 Interface



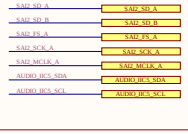
EXTERNAL HSE CLK



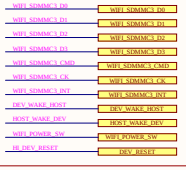
EXTERNAL LSE CLK



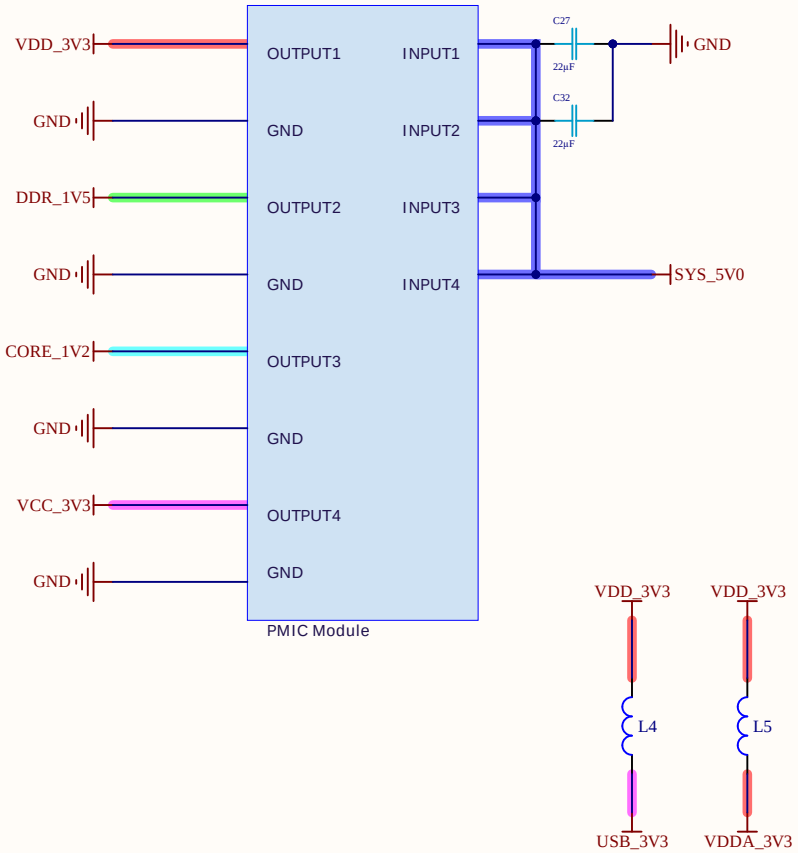
Audio



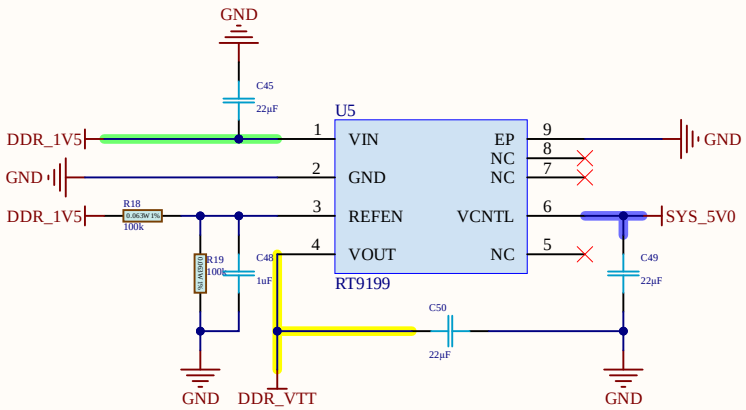
Wifi



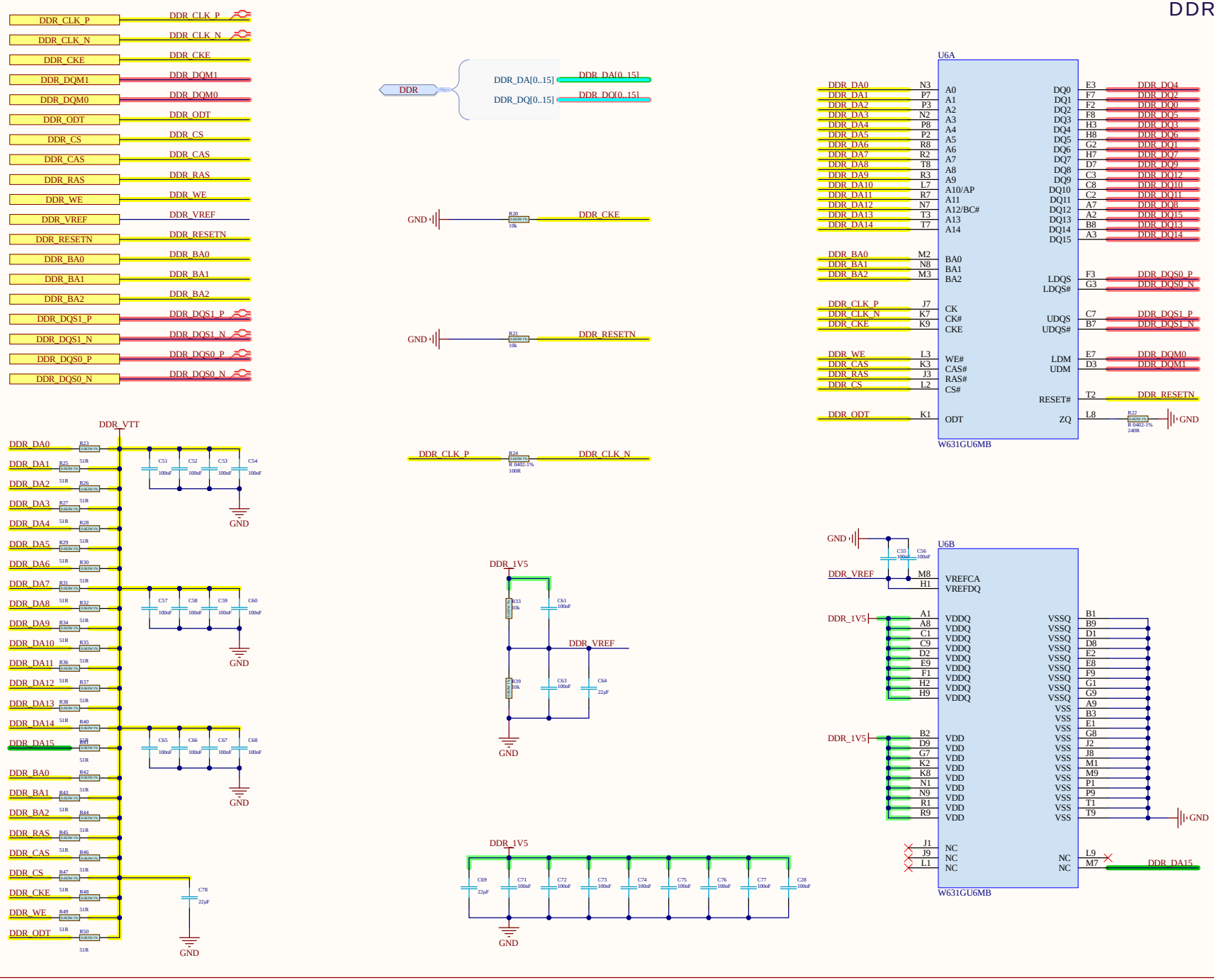
DC-DC


VDD_3V3
GND
DDR_1V5
GND
CORE_1V2
GND
VCC_3V3
GND
OUTPUT1
GND
OUTPUT2
GND
OUTPUT3
GND
OUTPUT4
GND
INPUT1
INPUT2
INPUT3
INPUT4
PMIC Module
C27
22µF
C32
22µF
GND
SYS_5V0
VDD_3V3
VDD_3V3
L4
L5
USB_3V3
VDDA_3V3

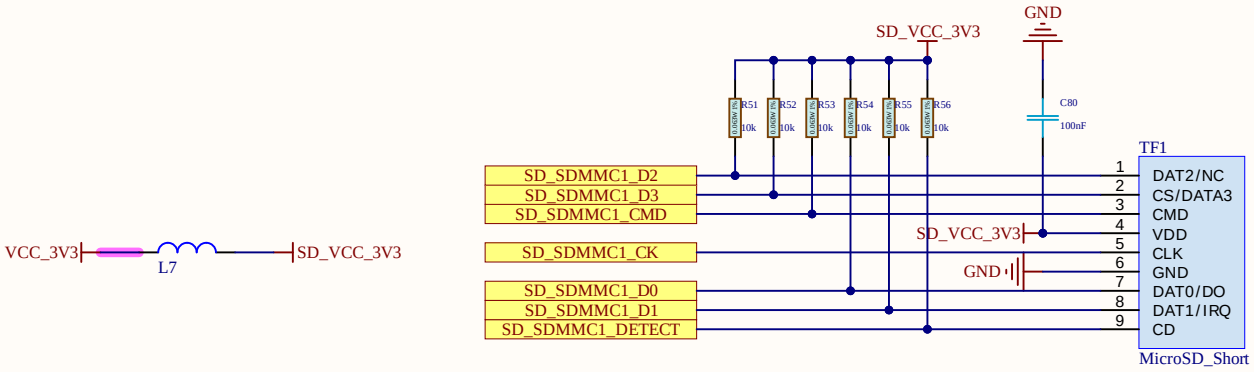
DDR VTT


GND
C45
22µF
U5
DDR_1V5
GND
DDR_1V5
R18
0.063W 1%
100k
R19
100k
C48
1uF
1
2
3
4
VIN
GND
REFEN
VOUT
EP
NC
NC
VCNTL
NC
9
8
7
6
5
RT9199
GND
SYS_5V0
C49
22µF
C50
22µF
GND
DDR_VTT
GND

DDR Module

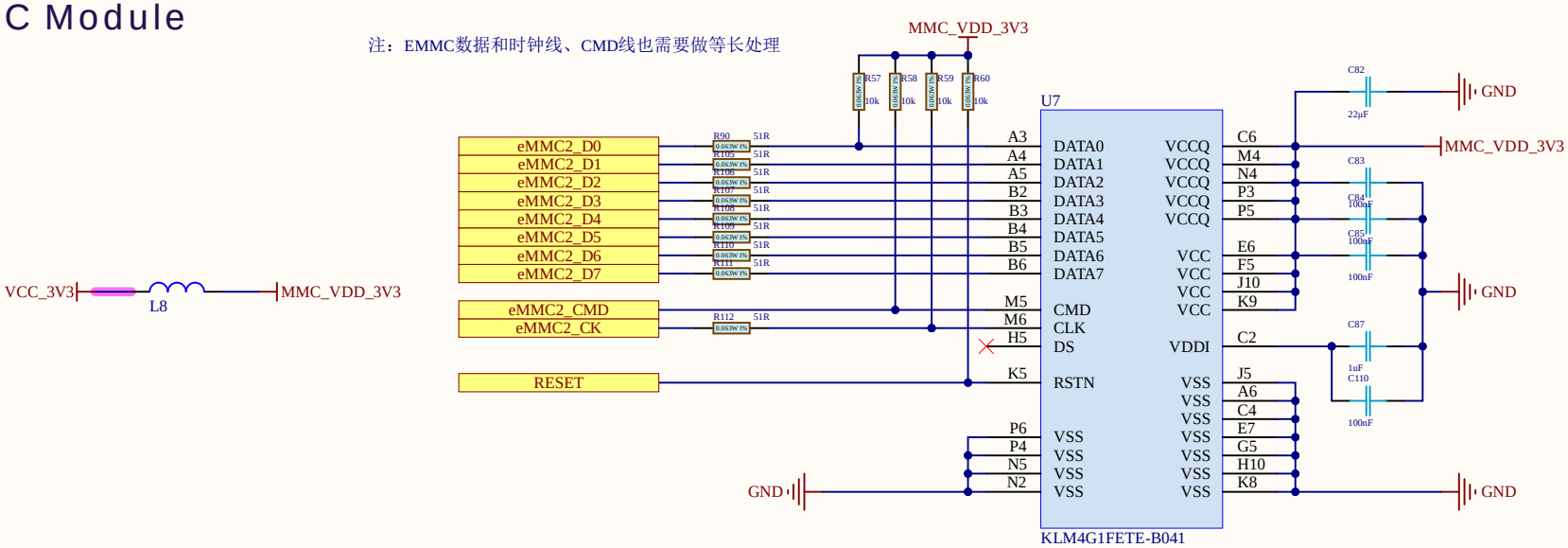

DDR_CLK_P
DDR_CLK_N
DDR_CKE
DDR_DQM1
DDR_DQM0
DDR_ODT
DDR_CS
DDR_CAS
DDR_RAS
DDR_WE
DDR_VREF
DDR_RESETN
DDR_BA0
DDR_BA1
DDR_BA2
DDR_DQS1_P
DDR_DQS1_N
DDR_DQS0_P
DDR_DQS0_N
DDR
DDR_DA[0..15]
DDR_DQ[0..15]
DDR_DA[0..15]
DDR_DQ[0..15]
GND
R20
10k
DDR_CKE
GND
R21
10k
DDR_RESETN
DDR_CLK_P
R24
R 0402-1%
100R
DDR_CLK_N
U6A
W631GU6MB
U6B
W631GU6MB
R22
R 0402-1%
240R
DDR_VTT
DDR_1V5
DDR_VREF
R33
10k
R39
10k
C61
100nF
C63
100nF
C64
22μF
C69
22μF
C71
100nF
C72
100nF
C73
100nF
C74
100nF
C75
100nF
C76
100nF
C77
100nF
C28
100nF
C78
22μF
C55
100nF
C56
100nF
C51
C52
C53
C54
C57
C58
C59
C60
C65
C66
C67
C68
100nF
51R
R23
R25
R26
R27
R28
R29
R30
R31
R32
R34
R35
R36
R37
R38
R40
R41
R42
R43
R44
R45
R46
R47
R48
R49
R50
0.063W 1%

SD Card Module

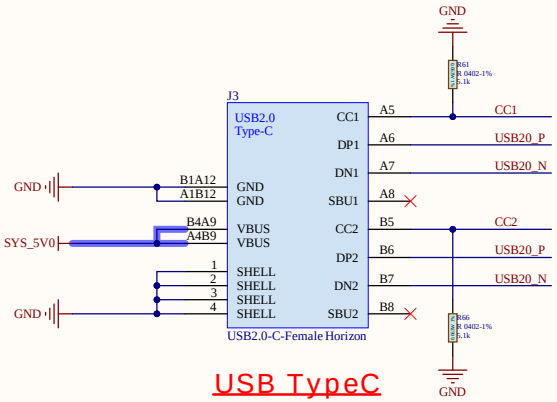


eMMC Module

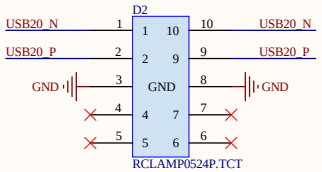
注：EMMC数据和时钟线、CMD线也需要做等长处理



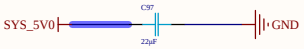
Type-C Power In



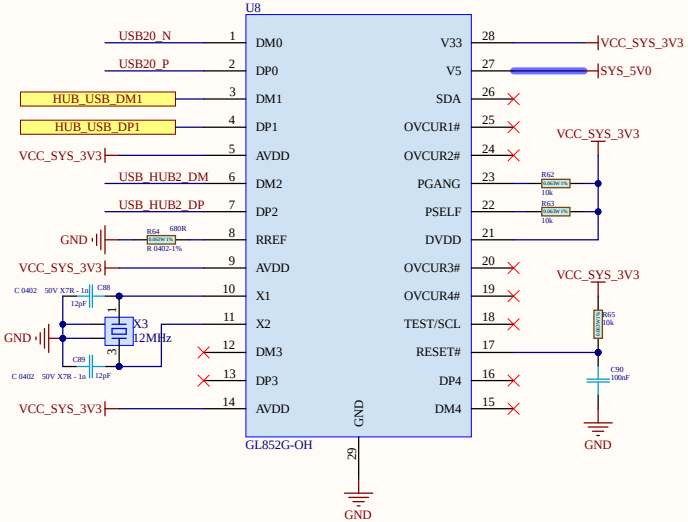
USB TypeC

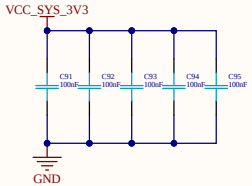


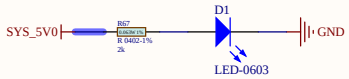
TVS



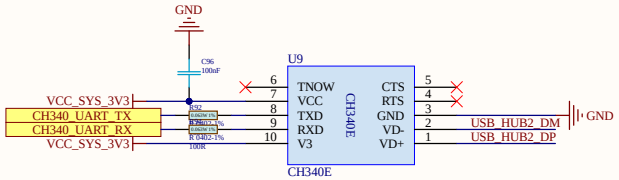
Type-C HUB



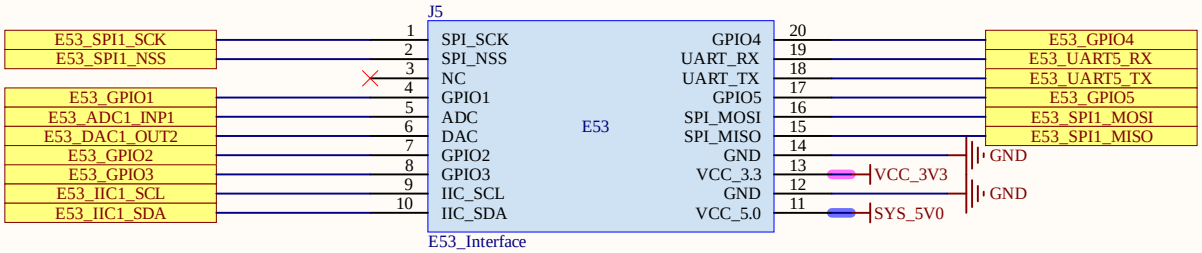


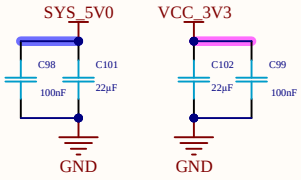


USB to UART

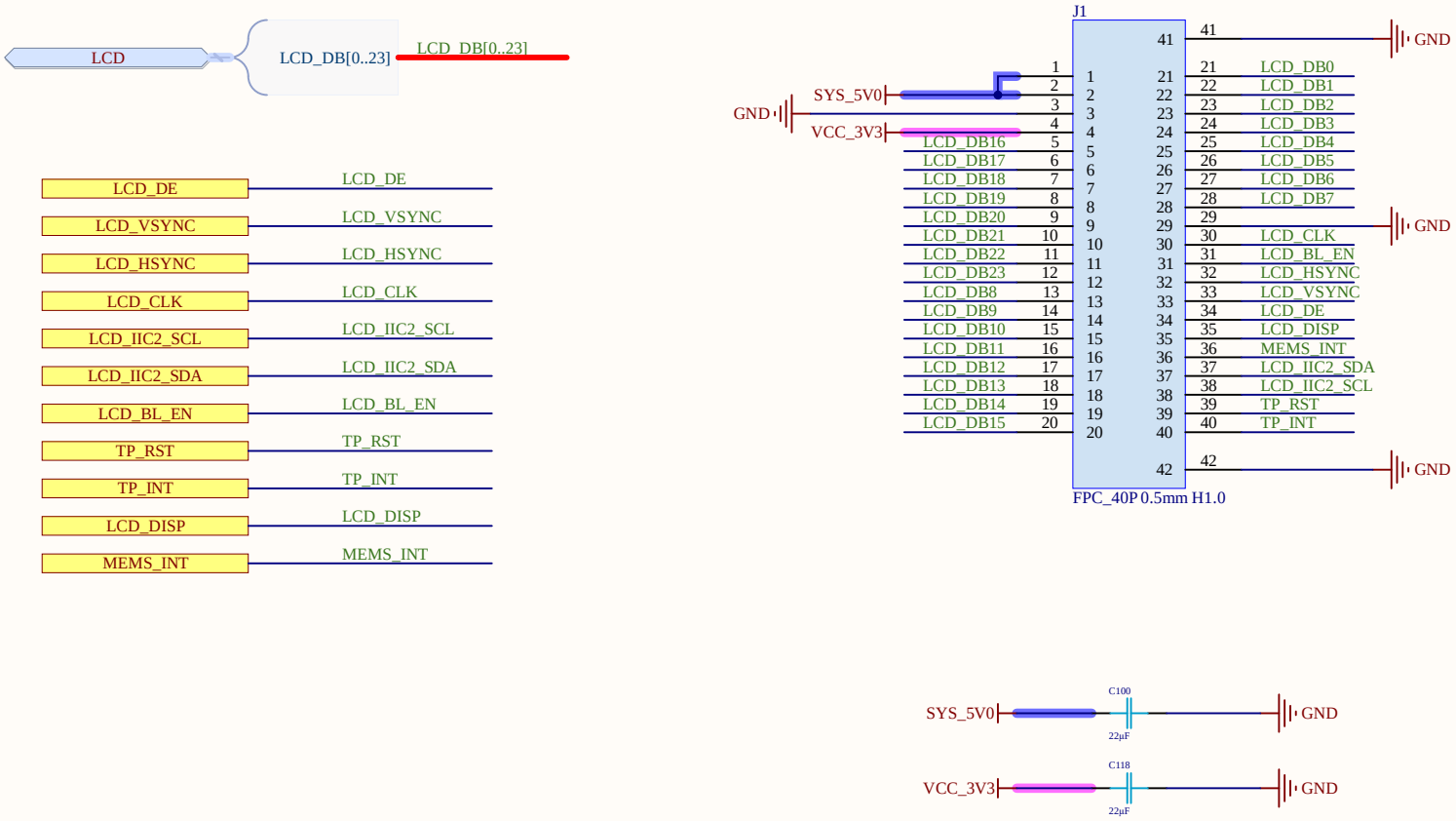


E53 Interface



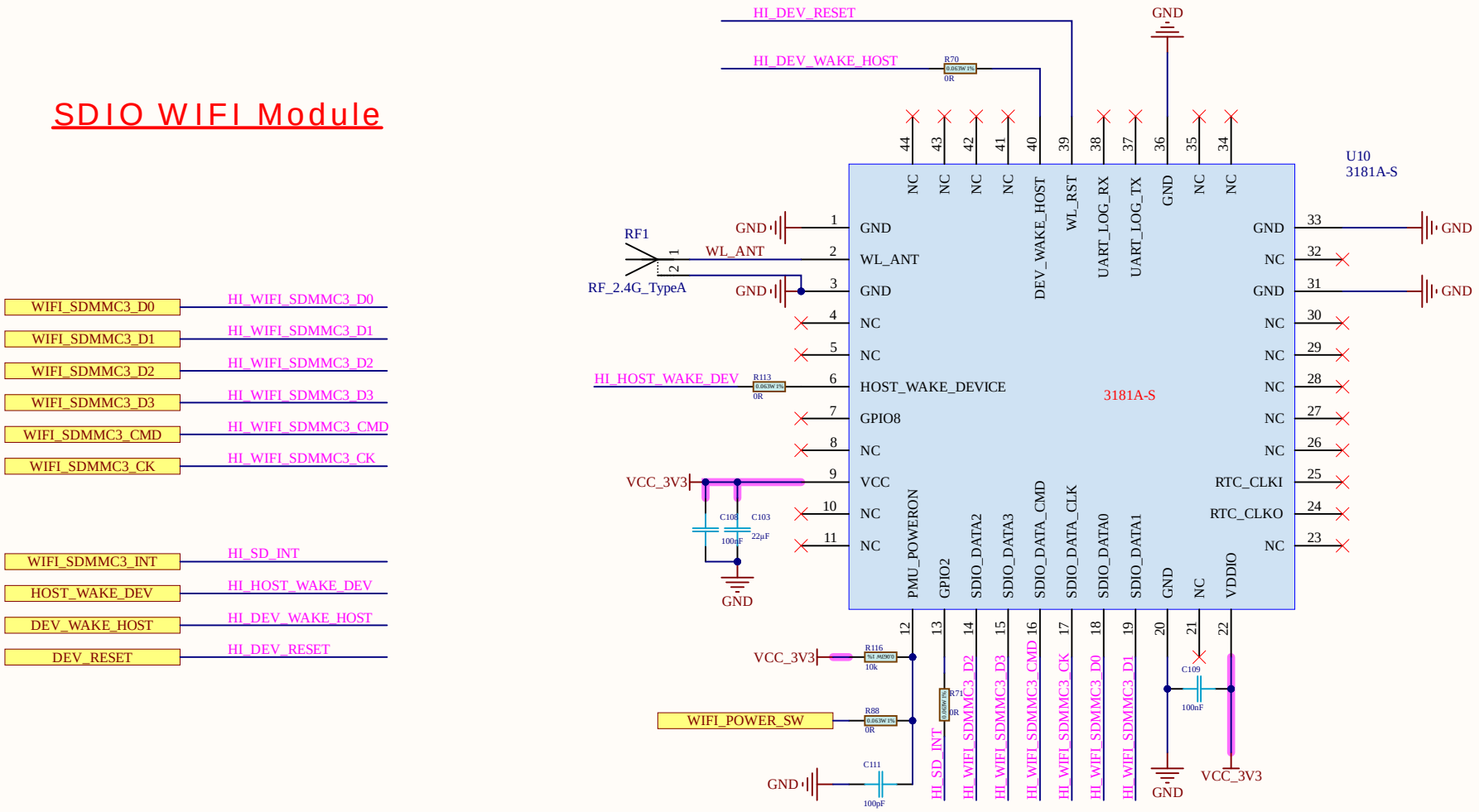


Display Module

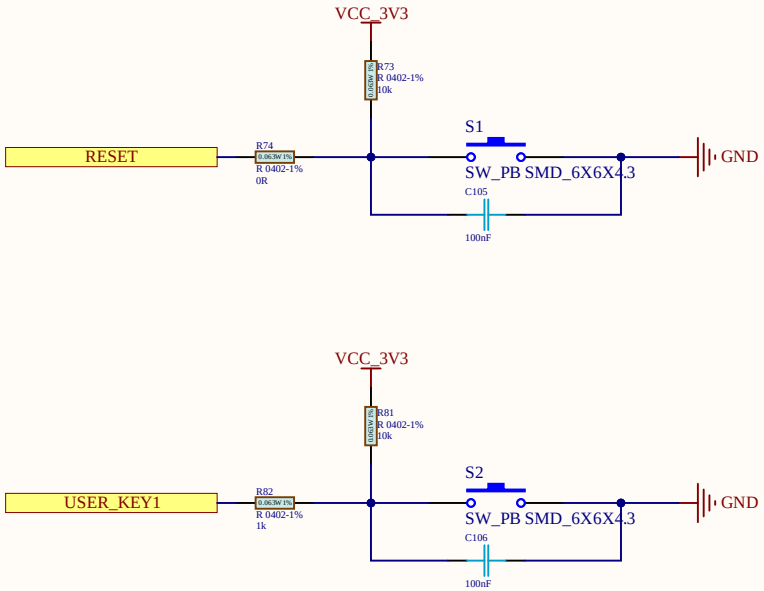


WiFi 3881

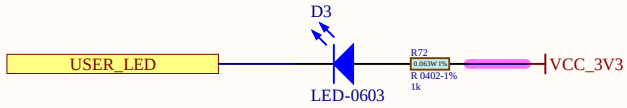
SDIO WIFI Module



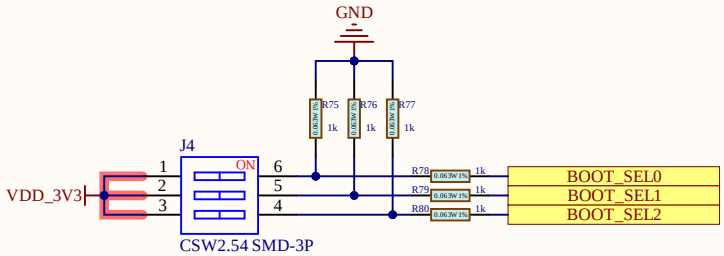
User's Key



User's LED

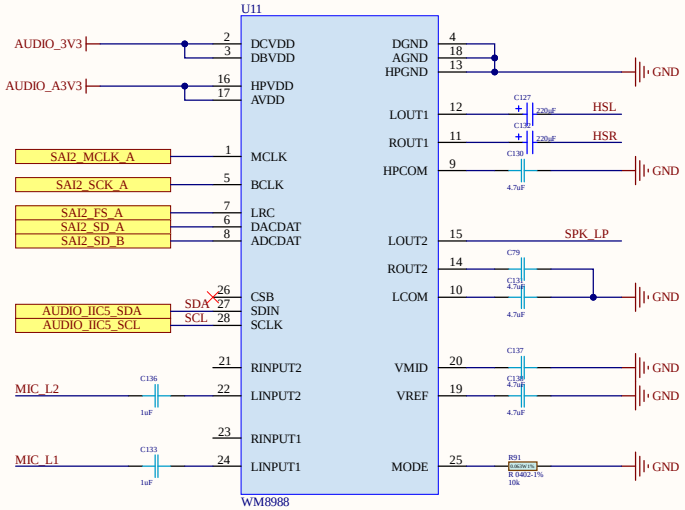


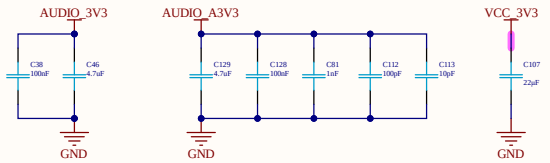
Boot

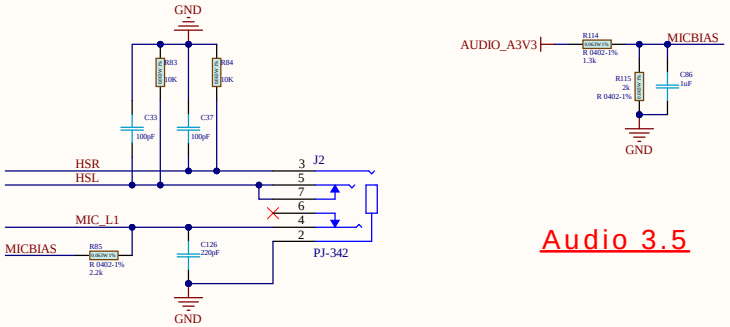


B2	B1	B0	BOOT Mode
0	0	0	UART and USB
0	1	0	eMMC
0	1	1	NAND Flash
1	0	1	SD Card
1	1	0	UART and USB

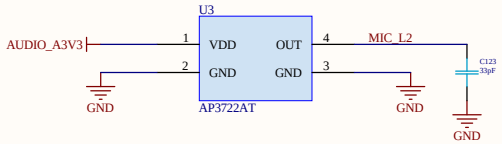
Audio Module





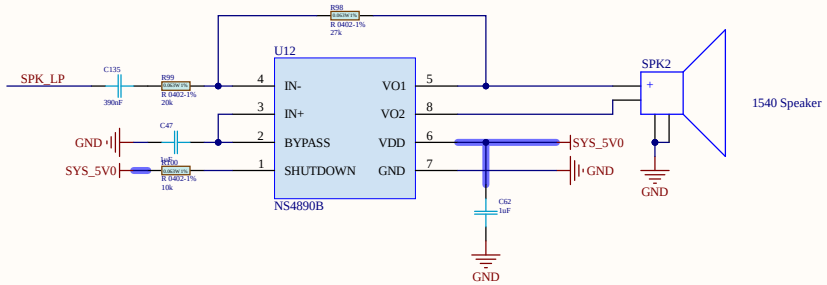


Audio 3.5

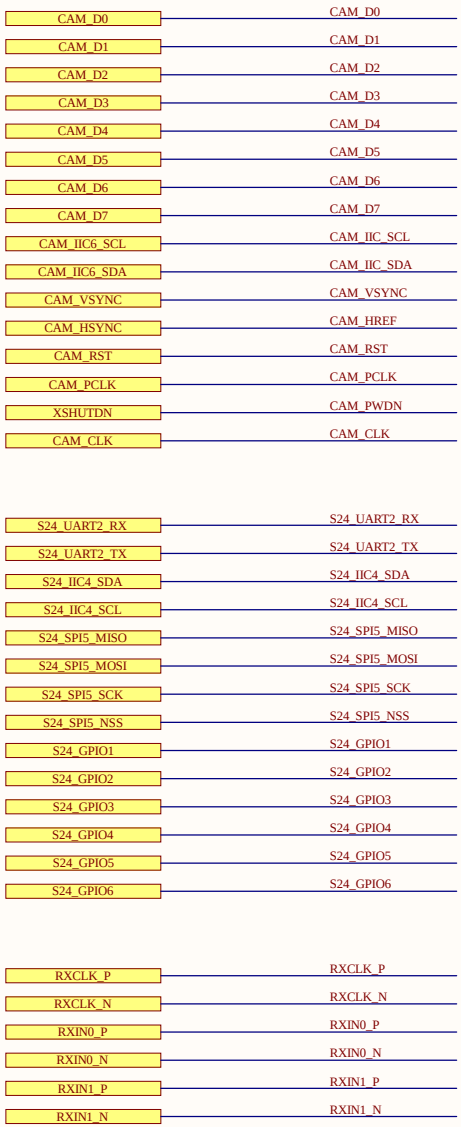


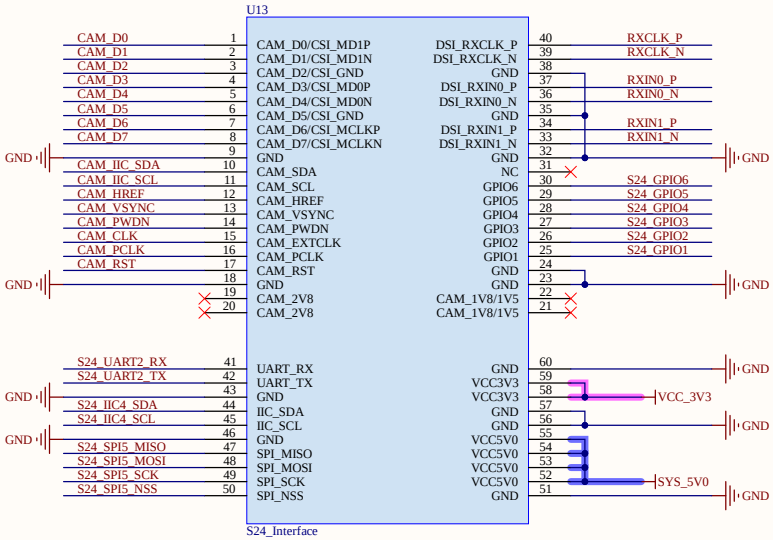
AUDIO IN

AUDIO OUT



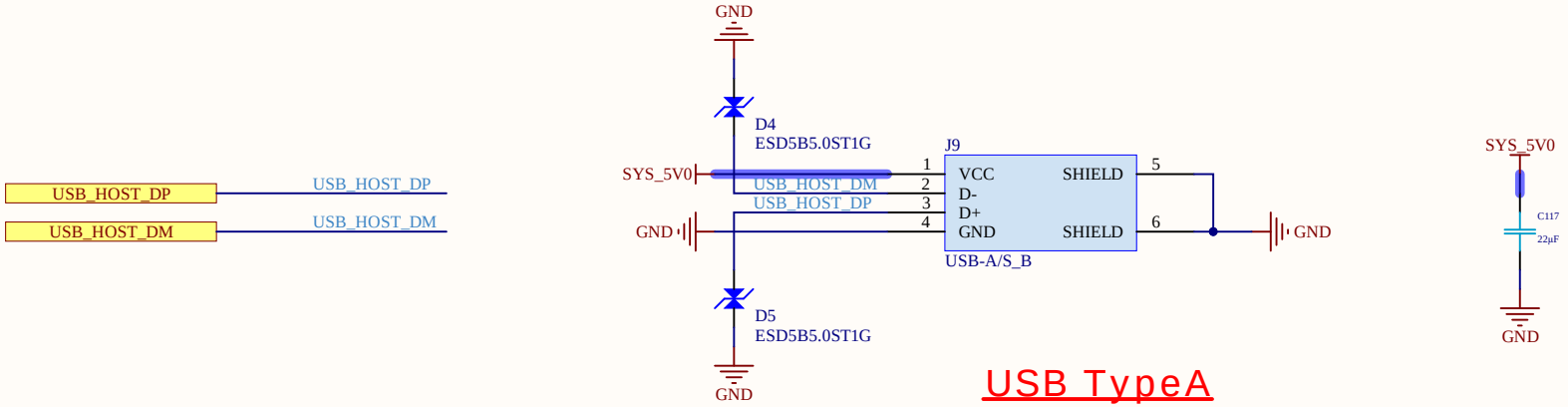
S24 Interface





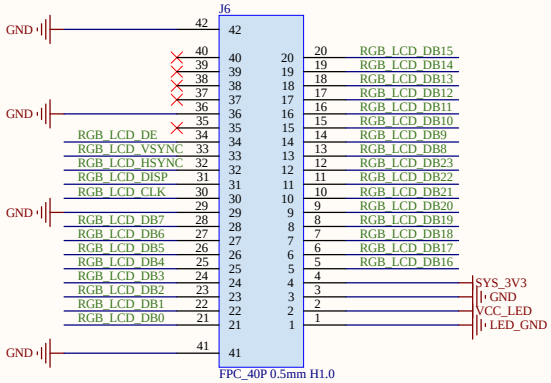
S24 Interface

USB HOST

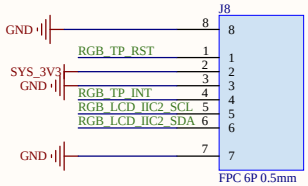


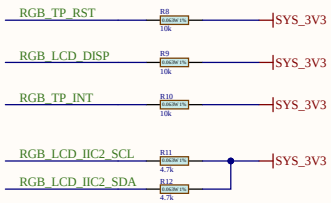
USB TypeA

Connector To 4.3' LCD



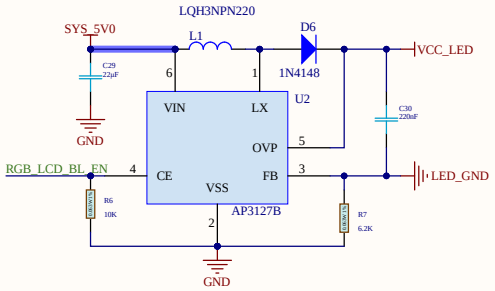
TOUCH



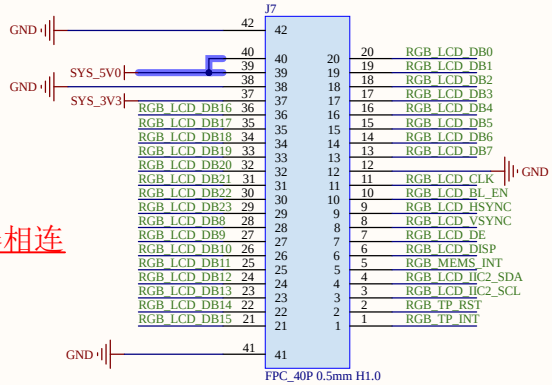


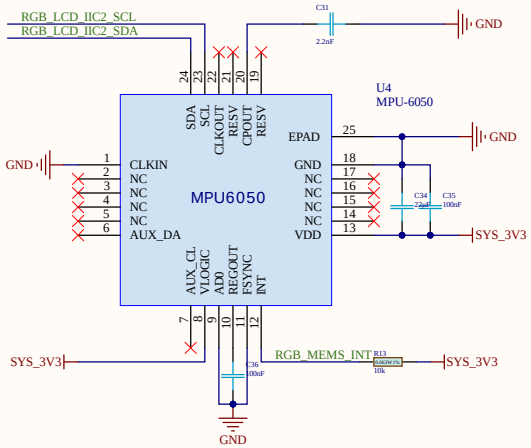
本页为屏幕板原理图

DC-DC



Connector To Board



通过软FPC线与主板J1连接器相连

MPU6050